

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks: 50

Annexure A

SCENARIO BASED QUESTIONS

Rubric Summary Sheet

Criteria	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total (10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S.CHANDRA SHEKHAR REDDY (Name), Reg No 192372163 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.CHANDRA SHEKHAR REDDY

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A retail analytics company is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks:

A. Handling Missing Values — i. Identify missing values in Age, Monthly_Spend, and Purchase_Frequency. ii. Apply at least two imputation techniques (e.g., Median Imputation, k-NN Imputation). iii. Compute the imputed values and justify the selected method.

B. Outlier Detection and Treatment — i. Detect outliers in Age and Monthly_Spend using the Z-score method. ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500). iii. Apply suitable treatment methods (capping / transformation / removal) and justify.

C. Categorical Data Standardization — i. Identify inconsistencies in Gender (M, Male, male, FEMALE, etc.). ii. Convert all entries into a standardized format (Male / Female).

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An e-commerce platform is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks:

A. Covariance Analysis — i. Compute covariance between Income and Purchase_Value (show steps or partial calculation). ii. Interpret whether the relationship is positive or negative. iii. Explain how covariance helps in identifying feature relationships.

B. Feature Selection — i. Identify redundant attributes using correlation or domain knowledge. ii. Select an optimal subset of features for modeling. iii. Justify your selection.

C. Data Transformation — i. Standardize the dataset (excluding Cust_ID and Customer_Class). ii. Explain why normalization/standardization is required.

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An education analytics team is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks:

A. Equal-Width Binning — i. Apply 3 bins for Study_Hours and Test_Score. ii. Calculate bin width and define intervals.

B. Equal-Frequency Binning — i. Divide data into 3 bins with equal records. ii. Assign values accordingly.

C. Concept Hierarchy Construction — Study_Hours → Low / Medium / High; Test_Score → Poor / Average / Excellent.

Q4. Normalization

Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

Q5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An e-commerce company is analyzing customer engagement by measuring the time spent (in seconds) on product pages. The duration varies significantly — from a few seconds to more than an hour — resulting in highly skewed data. To improve the effectiveness of customer segmentation using clustering techniques, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30

C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks:

A. Equal-Frequency Binning — i. Divide the dataset into 3 bins with approximately equal number of records. ii. Assign each value to its respective bin. iii. Analyze how effectively this method groups short and long visit durations.

B. Z-Score Normalization — i. Compute the mean and standard deviation of visit duration. ii. Apply Z-score normalization: $Z = (X - \mu) / \sigma$. iii. Discuss how this method handles extreme values (e.g., 1800, 3600 seconds).

C. Min-Max Normalization — i. Apply Min-Max normalization using range [0,1]: $X' = (X - X_{min}) / (X_{max} - X_{min})$. ii. Transform all values accordingly. iii. Compare results with Z-score normalization in terms of scaling and sensitivity to outliers.

ANSWERS

Annexure A: Answers to Scenario-Based Questions (CO2)

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

i. Missing values identified:

- Age → missing for C2
- Monthly_Spend (₹K) → missing for C3
- Purchase_Frequency → missing for C5

ii & iii. Two imputation techniques applied:

Method 1: Median Imputation

Median is preferred over mean because the dataset contains extreme outliers (Age = 105, Monthly_Spend = 500), and median is robust to such extremes.

Missing value	Sorted valid values used	Median	Imputed value
Age (C2)	23, 27, 31, 36, 40, 42, 105	36	36
Monthly_Spend (C3)	20, 25, 35, 38, 45, 300, 500	38	38
Purchase_Frequency (C5)	5, 8, 9, 10, 12, 45, 50	10	10

Method 2: k-NN Imputation (k = 3)

Nearest neighbours were chosen using Euclidean distance on the other numeric attributes (excluding records that themselves had missing values), then the target attribute was averaged over the 3 nearest neighbours.

Missing value	Basis (features used)	3 nearest neighbours	k-NN imputed value
Age (C2)	Monthly_Spend & Purchase_Frequency (35,8)	C7(36), C6(40), C1(23)	≈ 33
Monthly_Spend (C3)	Age & Purchase_Frequency (31,10)	C7(38), C6(45), C1(20)	≈ 34.3
Purchase_Frequency (C5)	Age & Monthly_Spend (27,25)	C1(5), C7(9), C6(12)	≈ 8.7 (≈9)

Justification: Median imputation is simple and resistant to the extreme values already present in the data (Age=105, Spend=500), so it will not be pulled off-centre the way a mean would be. k-NN imputation is more context-aware — it exploits the correlation between Monthly_Spend, Purchase_Frequency and Age (customers who spend more tend to purchase more often) and therefore gives a more personalised estimate than a single global statistic. In practice the k-NN estimates (33, 34.3, 8.7) are close to the medians (36, 38, 10), which cross-validates both methods; k-NN is preferred for the final imputation since it preserves inter-attribute relationships needed for the downstream recommendation system.

B. Outlier Detection and Treatment

i & ii. Z-score method: $Z = (X - \mu) / \sigma$, computed on the 8 records (using the median-imputed values above).

Attribute	Mean (μ)	Std Dev (σ)	Extreme record	Z-score
Age	42.5	24.37	C4 = 105	2.56 → Outlier
Monthly_Spend	125.13	166.55	C4 = 500	2.25 → Outlier
Monthly_Spend	125.13	166.55	C8 = 300	1.05 → borderline high, not a statistical outlier

Using a $|Z| > 2$ threshold (appropriate for this small sample), C4 (Age = 105, Spend = 500) is flagged as an outlier on both attributes; C8 (Spend = 300) is high but does not cross the threshold.

iii. Treatment:

- Age = 105 (C4): biologically implausible for a retail dataset and most likely a data-entry error (e.g. an extra digit). Treatment: correction/imputation — replace with the attribute median (36) or a value re-verified from source rather than deletion, to avoid losing the rest of C4's valid data.
- Monthly_Spend outliers (C4 = 500, C8 = 300): both belong to 'High' loyalty customers, so they are plausible genuine high-value spenders rather than errors. Treatment: capping/Winsorization (e.g. cap at the 95th percentile value) or a log-transformation, instead of removal, so the valuable high-spend signal is retained but its influence on distance-based algorithms (used later in the recommender) is reduced.

Justification: Removal would shrink an already small dataset and discard potentially useful 'High-loyalty' customer information; capping/transformation retains all 8 records while limiting the leverage of extreme values on models such as k-NN or clustering that are sensitive to scale.

C. Categorical Data Standardization

i. Inconsistent Gender entries found: M, Male, male (→ Male) and F, Female, FEMALE (→ Female).

ii. Standardized values:

Customer_ID	Original Gender	Standardized Gender
C1	M	Male
C2	Male	Male
C3	F	Female
C4	Female	Female
C5	male	Male
C6	FEMALE	Female
C7	M	Male
C8	Female	Female

Approach: convert all text to a single case, map abbreviations (M/F) to full labels, and apply a controlled vocabulary {Male, Female} using string standardization / mapping rules.

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis (Income vs Purchase_Value)

Mean Income = 80.625 (₹K), Mean Purchase_Value = 198.75 (₹K)

$$\text{Cov}(X,Y) = \Sigma(X_i - \bar{X})(Y_i - \bar{Y}) / n$$

User	Income dev.	Purchase_Value dev.	Product
U1	-40.625	-98.75	4011.72
U2	-20.625	-48.75	1005.47
U3	4.375	11.25	49.22
U4	39.375	121.25	4774.22
U5	-35.625	-88.75	3161.72
U6	-5.625	-18.75	105.47
U7	69.375	161.25	11186.72
U8	-10.625	-38.75	411.72

Sum of products $\approx 24,706.26 \rightarrow$ Covariance (population) = $24706.26 / 8 \approx 3088.3$ (₹K)²

ii. Interpretation: The covariance is strongly positive, so Income and Purchase_Value move together — customers with higher income tend to make higher-value purchases.

iii. Covariance helps identify feature relationships by quantifying the direction (and, once scaled to correlation, the strength) of the joint variation between two attributes. A large positive/negative covariance signals redundancy — the two features carry overlapping information — which is exactly the case for Income and Purchase_Value here.

B. Feature Selection

- i. Redundant attributes: Purchase_Value, Credit_Score, Avg_Balance and EMI all rise together with Income and with Customer_Class, indicating high mutual correlation and redundancy. Orders also correlates strongly with Purchase_Value.
- ii. Optimal subset for modeling: Income (overall financial capacity), Orders (behavioural engagement), and Credit_Score (creditworthiness — a comparatively independent risk dimension).
- iii. Justification: Income, Orders and Credit_Score each capture a distinct dimension of the customer (spending capacity, engagement frequency, credit risk), while Purchase_Value, Avg_Balance and EMI are largely explained by Income and would add multicollinearity without new information. Dropping them reduces computational cost and overfitting risk while preserving predictive power.

C. Data Transformation — Standardization (Z-scores)

$Z = (X - \mu) / \sigma$, applied to every numeric feature except Cust_ID and Customer_Class.

Feature			Mean		Std Dev		
Age			39.375		10.22		
Income			80.625		35.13		
Purchase_Value			198.75		88.52		
Credit_Score			723.125		54.25		
Orders			25.25		8.04		
Avg_Balance			25.125		12.36		
EMI			9500		4308.4		
User	Age	Income	Purch_Val ue	Credit_Scor e	Orders	Avg_Balanc e	EMI
U1	-1.51	-1.16	-1.12	-1.35	-1.28	-1.06	-1.04
U2	-0.53	-0.59	-0.55	-0.43	-0.65	-0.58	-0.58
U3	0.45	0.13	0.13	-0.06	0.34	-0.01	-0.12
U4	1.24	1.12	1.37	1.42	1.21	1.20	1.28
U5	-1.02	-1.01	-1.00	-0.80	-0.90	-0.90	-0.93
U6	-0.04	-0.16	-0.21	-0.24	-0.16	-0.25	-0.23
U7	1.63	1.98	1.82	1.79	1.83	2.01	1.97
U8	-0.23	-0.30	-0.44	-0.33	-0.40	-0.42	-0.35

- ii. Normalization/standardization is required because the features are on very different scales and units (Age in years vs EMI in thousands of rupees). Without scaling, distance- and gradient-based algorithms (k-NN, clustering, PCA, regression) would be dominated by the large-magnitude features (e.g. EMI, Income) purely due to scale, not actual importance. Standardizing to mean 0 / std 1 ensures every feature contributes comparably to the model.

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning (3 bins)

Study_Hours: min = 2, max = 11, range = 9, bin width = $9/3 = 3$

Bin	Interval	Values in bin
Bin 1 (Low)	[2 – 5)	2, 3, 4
Bin 2 (Medium)	[5 – 8)	5, 6, 7
Bin 3 (High)	[8 – 11]	8, 9, 10, 11

Test_Score: min = 40, max = 100, range = 60, bin width = $60/3 = 20$

Bin	Interval	Values in bin
Bin 1 (Low)	[40 – 60)	40, 50, 55
Bin 2 (Medium)	[60 – 80)	65, 75

Bin 3 (High)	[80 – 100]	85, 90, 95, 98, 100
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B. Equal-Frequency Binning (3 bins, equal records)

10 records split ~evenly as 4-3-3, in sorted order:

Bin	Students	Study_Hours	Test_Score
Bin 1	S1–S4	2, 3, 4, 5	40, 50, 55, 65
Bin 2	S5–S7	6, 7, 8	75, 85, 90
Bin 3	S8–S10	9, 10, 11	95, 98, 100

C. Concept Hierarchy Construction

Attribute	Low/Poor	Medium/Average	High/Excellent
Study_Hours	2 – 5 hrs (Low)	6 – 8 hrs (Medium)	9 – 11 hrs (High)
Test_Score	40 – 59 (Poor)	60 – 79 (Average)	80 – 100 (Excellent)

Q4. Normalization of 200, 300, 400, 600, 1000

(a) Min-Max Normalization [0,1]

$X' = (X - \min) / (\max - \min)$, min = 200, max = 1000, range = 800

X	Calculation	X' (Min-Max)
200	(200-200)/800	0.000
300	(300-200)/800	0.125
400	(400-200)/800	0.250
600	(600-200)/800	0.500
1000	(1000-200)/800	1.000

(b) Z-score Normalization

Mean $\mu = 500$, Std Dev $\sigma = 282.84$ (population)

X	$X - \mu$	$Z = (X - \mu) / \sigma$
200	-300	-1.061
300	-200	-0.707
400	-100	-0.354
600	100	0.354
1000	500	1.768

Q5. Data Transformation in Online Shopping Behavior Analysis

A. Equal-Frequency Binning (3 bins)

Bin	Customers	Visit_Duration (s)	Interpretation
Bin 1 – Short visits	C1–C4	15, 30, 45, 60	Quick, bounce-type visits
Bin 2 – Medium visits	C5–C7	120, 300, 600	Moderately engaged browsing
Bin 3 – Long visits	C8–C10	900, 1800, 3600	Highly engaged / deep-browsing customers

iii. Because equal-frequency binning fixes the number of records per bin rather than the value range, it copes well with the heavy right-skew of this data — each segment (short/medium/long) ends up with a comparable sample size for downstream clustering, even though Bin 3 spans a much wider numeric range than Bin 1.

B. Z-Score Normalization

Mean $\mu = 747$ s, Std Dev $\sigma \approx 1090.9$ s (population)

Customer	Visit_Duration	Z-score
C1	15	-0.671
C2	30	-0.657
C3	45	-0.644
C4	60	-0.630
C5	120	-0.575
C6	300	-0.410
C7	600	-0.135
C8	900	0.140
C9	1800	0.965
C10	3600	2.616

iii. Z-score normalization does not bound the outliers — 1800s and 3600s still produce visibly large positive z-scores (0.97 and 2.62). It centres and scales the data around the mean but the extreme values remain 'extreme' in the transformed space, which can still exert leverage on distance-based clustering, though far less than the raw values would.

C. Min-Max Normalization [0,1]

$X' = (X - \min)/(\max - \min)$, min = 15, max = 3600, range = 3585

Customer	Visit_Duration	X' (Min-Max)
C1	15	0.0000
C2	30	0.0042
C3	45	0.0084
C4	60	0.0126
C5	120	0.0293
C6	300	0.0795
C7	600	0.1632
C8	900	0.2469
C9	1800	0.4979
C10	3600	1.0000

iii. Comparison: Min-Max normalization is highly sensitive to the single outlier (3600s) — because it anchors the scale, all 'typical' short visits (15–120s) are compressed into a narrow band near 0 (0.000–0.029), losing discriminative power among ordinary customers. Z-score normalization is centred on the mean and scaled by the standard deviation, so it preserves relative distances among the bulk of the data better and is less distorted by a single extreme point, though the outliers still receive large z-scores. For this heavily skewed dataset, Z-score normalization (or a log-transform before scaling) is generally the better choice for clustering, since Min-Max would let one extreme session dominate the scale.